Chapter 12. Nonparametric Regression

Kernel regression, local polynomial regression, bandwidth and order selection, confidence intervals

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Model

$Y_i = m(X_i) + \sigma(X_i)\epsilon_i$ with $E\epsilon_i = 0$, $\operatorname{Var}\epsilon_i = 1$ and $m(x) = E(Y \mid X = x)$ smooth but otherwise unrestricted. The goal is to estimate the curve m and quantify its uncertainty.

12.1 Local polynomial regression

Motivation. To estimate a regression function without a parametric form, fit a low-degree polynomial in a window around each point, weighting nearby observations more. Local linear fits correct the boundary bias of local averages.

Definition (Local polynomial estimator)

With $K_h(u) = K(u/h)/h$, fit at each x

$$\hat{\boldsymbol{\beta}}_x = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \big(Y_i - \beta_0 - \beta_1(X_i - x) - \cdots - \beta_p(X_i - x)^p\big)^2 K_h(x - X_i),$$

and set $\hat{m}_p(x) = \hat{\beta}_{0,x}$. In matrix form, $\hat{\boldsymbol{\beta}}_x = (X_x' W_x X_x)^{-1} X_x' W_x \mathbf{y}$, with $W_x = \operatorname{diag}(K_h(x - X_i))$ and X_x having rows $(1, X_i - x, \ldots, (X_i - x)^p)$.

Every local polynomial estimator is linear in the responses: $\hat{m}_p(x) = \sum_i w_i^p(x) Y_i$.

Proposition 12.1

With $S_q = \sum_j K_h(x - X_j)(X_j - x)^q$:

(i) $p = 0$ (**Nadaraya–Watson**): $w_i^0(x) = K_h(x - X_i) / \sum_j K_h(x - X_j)$;

(ii) $p = 1$ (**local linear**): $w_i^1(x) = \dfrac{\{S_2 - (X_i - x)S_1\} K_h(x - X_i)}{S_0 S_2 - S_1^2}$.

In both cases $\sum_i w_i(x) = 1$. For $p = 1$ also $\sum_i w_i^1(x)(X_i - x) = 0$: the fit reproduces straight lines exactly.

Proof [Supplement]. (i) Minimize $\sum (Y_i - \beta_0)^2 K_h(x - X_i)$.

(ii) $X_x' W_x X_x = \begin{pmatrix} S_0 & S_1 \\ S_1 & S_2 \end{pmatrix}$, and the first row of its inverse is $(S_2, -S_1)/(S_0 S_2 - S_1^2)$. Multiplying by $X_x' W_x$ gives the weights. Summing, $\sum_i (S_2 - (X_i - x)S_1) K_h = S_2 S_0 - S_1^2$, and $\sum_i (S_2 - (X_i - x)S_1)(X_i - x) K_h = S_2 S_1 - S_1 S_2 = 0$. ∎

As $h \to \infty$ the Nadaraya–Watson estimator tends to the constant $\bar{Y}$, while the local linear estimator tends to the least squares line.

Theorem 12.2

Let m be twice continuously differentiable, let the design density f be smooth with $f(x) > 0$, and use the Gaussian kernel. As $n \to \infty$, $h \to 0$ and $nh \to \infty$, conditionally on the X's and for $p = 0, 1$,

$$\operatorname{bias} \hat{m}_p(x) = B_p(x) h^2 + o_P(h^2), \qquad \operatorname{Var} \hat{m}_p(x) = \frac{\sigma^2(x)}{2\sqrt{\pi} f(x)} \cdot \frac{1}{nh} + o_P\Big(\frac{1}{nh}\Big),$$

with $B_0 = \dfrac{m''}{2} + \dfrac{m'f'}{f}$ and $B_1 = \dfrac{m''}{2}$.

Sketch. Since the weights sum to 1, bias $= \sum_i w_i(x)\{m(X_i) - m(x)\}$ and $\mathrm{Var} = \sum_i w_i^2 \sigma^2(X_i)$. For $p = 0$, the numerator $\frac{1}{n}\sum K_h(x - X_i)\{m(X_i) - m(x)\}$ has mean $\int K(u)\{m(x - hu) - m(x)\}f(x - hu)\,du = h^2(\frac{m''f}{2} + m'f') + o(h^2)$. The denominator tends to $f(x)$, giving B_0. For the variance, $\frac{1}{n^2}\sum K_h^2\sigma^2 \approx \frac{\sigma^2 f}{nh}R(K)$ with $R(\phi) = \frac{1}{2\sqrt{\pi}}$; divide by f^2. For $p = 1$, the property $\sum w_i^1(X_i - x) = 0$ removes the first-order term $m'(x)(X_i - x)$ exactly, including the asymmetry of the design that produced $m'f'/f$. The leading variance is unchanged. ■

The local linear estimator has the same leading variance as Nadaraya–Watson but no **design bias** $m'f'/f$. It also behaves better near the boundary, which is why it is the default. Generally the bias decreases with the degree, $|\mathrm{bias}\,\hat{m}_0| \gtrsim |\mathrm{bias}\,\hat{m}_1| \gtrsim |\mathrm{bias}\,\hat{m}_2| \gtrsim \dots$. The leading variance follows the pattern

$$\mathrm{Var}\,\hat{m}_0 \approx \mathrm{Var}\,\hat{m}_1 < \mathrm{Var}\,\hat{m}_2 \approx \mathrm{Var}\,\hat{m}_3 < \cdots .$$

Raising an even degree to the next odd one therefore reduces bias at no cost in variance, which is why odd degrees (usually $p = 1$, sometimes $p = 3$) are preferred. The bias $\frac{m''}{2}h^2$ is positive in valleys and negative at peaks, so the estimate flattens features of the curve.

12.2 Choosing the bandwidth

Motivation. The bandwidth should minimize prediction error, which we cannot observe. Leave-one-out cross-validation estimates it, and for linear smoothers it can be computed from a single fit.

Integrating the bias and variance against the design density,

$$\mathrm{MISE} \approx \frac{h^4}{4}\int (m'')^2 f + \frac{\int \sigma^2}{2\sqrt{\pi}\,nh}, \qquad h^* = \Big(\frac{\int \sigma^2}{2\sqrt{\pi}\int (m'')^2 f \cdot n}\Big)^{1/5}.$$

- **Direct plug-in** (Ruppert–Sheather–Wand): estimate $\int (m'')^2 f$ and $\int \sigma^2$ by blockwise quartic fits, with the number of blocks chosen by an information criterion.
- **Leave-one-out cross-validation:** minimize $\mathrm{CV}(h) = \frac{1}{n}\sum_i \{Y_i - \hat{m}^{(-i)}(X_i)\}^2$.

Proposition 12.3

For a linear smoother whose leave-one-out fit equals the full fit at X_i after deleting point i (true for local polynomial fits), $\mathrm{CV}(h) = \dfrac{1}{n}\sum_i \Big(\dfrac{Y_i - \hat{m}(X_i)}{1 - w_i(X_i)}\Big)^2$.

Proof [Supplement]. Let $\tilde{Y}_i = \hat{m}^{(-i)}(X_i)$. Replacing Y_i by $\tilde{Y}_i$ leaves the weighted fit at X_i unchanged: the added point lies exactly on the leave-one-out local fit, so it does not change the weighted least squares solution. Hence $\tilde{Y}_i = \sum_{j \neq i} w_j(X_i)Y_j + w_i(X_i)\tilde{Y}_i$, so $\tilde{Y}_i = \frac{\hat{m}(X_i) - w_i(X_i)Y_i}{1 - w_i(X_i)}$ and $Y_i - \tilde{Y}_i = \frac{Y_i - \hat{m}(X_i)}{1 - w_i(X_i)}$. ■

So CV needs only one fit per h.

12.3 Confidence intervals

As for densities, intervals target the smoothed curve $m_h(x) = E\hat{m}_1(x)$.

- **Theory-based:** $\widehat{\mathrm{se}}(\hat{m}_1(x)) = \big(\sum_i w_i^1(x)^2\hat{\sigma}^2(X_i)\big)^{1/2}$ and $m_h(x) \in \hat{m}_1(x) \pm z_{\alpha/2}\widehat{\mathrm{se}}$. Under homoscedasticity, with W the $n \times n$ smoother matrix ($\hat{\mathbf{y}} = W\mathbf{y}$),

$$\hat{\sigma}^2 = \frac{\|(I-W)\mathbf{y}\|^2}{\mathrm{tr}\{(I-W)'(I-W)\}}.$$

  Under heteroscedasticity, smooth $\log r_i^2$ against X_i and exponentiate, which keeps $\hat{\sigma}^2(x) > 0$.
- **Bootstrap:** resample pairs (X_i, Y_i), refit with the *same* h on a grid, and use $\hat{m}_1(x) \pm z_{\alpha/2}\widehat{\mathrm{se}}^*(x)$ or the percentile interval $(Q^*_{\alpha/2}(x), Q^*_{1-\alpha/2}(x))$. Residual bootstraps need care because the residuals inherit the smoothing bias.

Simultaneous bands are possible, as in Section 11.4, but are rarely used in practice.

12.4 Selected exercises

Motivation. The exercises identify the parametric limits of the two smoothers, show that local linear fitting is exactly unbiased for straight lines, and use this to explain when the residual variance estimate is unbiased.

Exercise 12.1, 12.4 (Infinite bandwidth)

Let K be continuous at 0 with $K(0) > 0$ (e.g. Gaussian), and suppose the X_i are not all equal. Show that as $h \to \infty$, for every x:
(a) the Nadaraya–Watson estimate tends to $\bar{Y}$;
(b) the local linear estimate tends to the least squares line $\bar{Y} + \hat{\beta}(x - \bar{X})$, $\hat{\beta} = S_{XY}/S_{XX}$.

Solution.

Step 1. (Scale invariance.) In Proposition 12.1, w_i^0 is a ratio of expressions of degree 1 in the kernel values $K_h(x - X_j)$, and w_i^1 a ratio of expressions of degree 2. So both are unchanged if all $K_h(x - X_j)$ are multiplied by a common constant. Replace $K_h(x - X_j)$ by $\kappa_j(h) = K((x - X_j)/h)$; then $\kappa_j(h) \to K(0) > 0$ as $h \to \infty$.

(a)

Step 2. $w_i^0 = \kappa_i / \sum_j \kappa_j \to K(0)/(nK(0)) = 1/n$, so $\hat{m}_0(x) \to \bar{Y}$.

(b)

Step 3. Let $d_j = X_j - x$. Then $S_q \to K(0)\sum_j d_j^q$, and the denominator tends to

$$K(0)^2\Big\{n\sum_j d_j^2 - \Big(\sum_j d_j\Big)^2\Big\} = K(0)^2\, nS_{XX} > 0.$$

So $w_i^1 \to (\sum_j d_j^2 - d_i\sum_j d_j)/(nS_{XX})$.

Step 4. Put $\delta = \bar{X} - x$, so $\sum_j d_j = n\delta$, $\sum_j d_j^2 = S_{XX} + n\delta^2$ and $\sum_i d_i Y_i = S_{XY} + n\delta\bar{Y}$. Then

$$\sum_i w_i^1 Y_i \to \frac{(S_{XX} + n\delta^2)\, n\bar{Y} - n\delta(S_{XY} + n\delta\bar{Y})}{nS_{XX}} = \bar{Y} - \delta\frac{S_{XY}}{S_{XX}} = \bar{Y} + \hat{\beta}(x - \bar{X}).$$

So the global least squares line is the large-bandwidth limit of the local linear fit, while the Nadaraya–Watson fit flattens to the constant $\bar{Y}$. ■

Exercise 12.5 (No bias for straight lines)

Suppose $m(x) = \alpha + \beta x$. Show that, conditionally on the X's, the local linear estimator is exactly unbiased for every h and x, while the Nadaraya–Watson estimator has bias $\beta \sum_i w_i^0(x)(X_i - x)$, which is nonzero in general (e.g. at $x = \min_i X_i$). Relate this to B_0 and B_1 in Theorem 12.2.

Solution.

Step 1. Since $E(Y_i \mid X) = m(X_i)$ and the weights depend only on the X's,

$$E(\hat{m}_p(x) \mid X) - m(x) = \sum_i w_i^p(x) m(X_i) - m(x) = \beta \sum_i w_i^p(x)(X_i - x),$$

using $\sum_i w_i^p = 1$ and $m(X_i) - m(x) = \beta(X_i - x)$.

Step 2. For $p = 1$, $\sum_i w_i^1(x)(X_i - x) = 0$ by Proposition 12.1, so the bias is exactly 0.

Step 3. For $p = 0$ the bias is $\beta \sum_i K_h(x - X_i)(X_i - x) / \sum_i K_h(x - X_i)$. At $x = \min_i X_i$ every $X_i - x \geq 0$, with at least one strictly positive (unless all X_i are equal). With positive kernel weights the bias then has the sign of β and is nonzero: the fit is pulled toward the interior.

This is consistent with Theorem 12.2: $B_1 = m''/2 = 0$ for a line, while $B_0 = m' f'/f = \beta f'/f$ is the design bias, nonzero wherever the design density is not flat. ■

Exercise 12.6 (When is $\hat{\sigma}^2$ unbiased?)

Let $\hat{\mathbf{y}} = W\mathbf{y}$ be a linear smoother with $E(\mathbf{y} \mid X) = \mathbf{m} = (m(X_i))_i$ and $\mathrm{Var}(\mathbf{y} \mid X) = \sigma^2 I$. Show that

$$E(\|(I - W)\mathbf{y}\|^2 \mid X) = \|(I - W)\mathbf{m}\|^2 + \sigma^2 \mathrm{tr}\{(I - W)'(I - W)\}.$$

Deduce that the estimator $\hat{\sigma}^2$ of Section 12.3 is unbiased when m is linear and W is the local linear smoother, and biased upward otherwise.

Solution [Supplement].

Step 1. Write $\mathbf{y} = \mathbf{m} + \boldsymbol{\epsilon}$ with $E\boldsymbol{\epsilon} = 0$ and $\mathrm{Var}\,\boldsymbol{\epsilon} = \sigma^2 I$, and put $A = (I - W)'(I - W)$. Then

$$\|(I - W)\mathbf{y}\|^2 = \|(I - W)\mathbf{m}\|^2 + 2\mathbf{m}' A\boldsymbol{\epsilon} + \boldsymbol{\epsilon}' A\boldsymbol{\epsilon}.$$

Step 2. The middle term has mean 0. For the last, $E\boldsymbol{\epsilon}' A\boldsymbol{\epsilon} = E\mathrm{tr}(A\boldsymbol{\epsilon}\boldsymbol{\epsilon}') = \mathrm{tr}(A\,\sigma^2 I) = \sigma^2 \mathrm{tr} A$. This proves the identity.

Step 3. Hence $E\hat{\sigma}^2 = \sigma^2 + \|(I - W)\mathbf{m}\|^2 / \mathrm{tr} A \geq \sigma^2$. Row i of W holds the weights $w_j(X_i)$ of the fit at X_i. If m is linear and W is local linear, Exercise 12.5 gives $\sum_j w_j(X_i) m(X_j) = m(X_i)$ for every i, i.e. $(I - W)\mathbf{m} = 0$, and $\hat{\sigma}^2$ is unbiased. Otherwise the smoothing bias enters $\hat{\sigma}^2$ through $\|(I - W)\mathbf{m}\|^2$ and inflates it. ■